← Action 10B

One greeting. Three environments.

15-minute lecture / Investigation 0

For every step: predict → try → observe → explain. Keep a short note and an evidence file/link, including during the lecture. If something fails, keep the error and your next step.

> **0-2 min / investigate the lecture**
>
> Write three concepts, predict one output and record one question. Keep a whiteboard/slide photo or annotated sketch as evidence. The lecture is Investigation 0.

> **2-4 min / type and test**
>
> Type short blocks, predict, run and read the result. Notice braces, parentheses and semicolons. Assistive input and recovery copying remain available; explain what the code does.

> **4-7 min / class, object and method**
>
> Greeting describes a type. new Greeting creates an object. Name stores data. BuildMessage is an instance method that returns a string. A variable of this class type holds an object reference.

> **7-10 min / same logic, different output**
>
> Console.WriteLine writes to the console. XAML describes WPF controls; a C# Click handler sets MessageText.Text. An engine script uses the engine's output/log or screen API.

> **10-13 min / events and lifecycle**
>
> A button calls a handler through its Click event. Unity calls Start; Godot calls _Ready. The base class and callback belong to the chosen engine. x:Name identifies a WPF control; it is not a data binding.

> **13-15 min / release to investigation**

> Whiteboard class → object → method → output. Work solo or choose a group with a team lead and task list. Attempt all three builds; keep notes and evidence as you go.

Concepts / when we use them

> **Lecture**
>
> Context, audience, three hats, prediction, notes and evidence

> **Console**
>
> Project/file structure, class, object, new, property, method, return, string interpolation

> **XAML / WPF**
>
> Window, layout, TextBlock, Button, x:Name, Click event, partial class, code-behind

> **Engine**
>
> Official research, editor/SDK, scene, object, script, inheritance, callback, debug/output

> **Submit**
>
> Investigation record, whiteboard/peer evidence, team lead/tasks if grouped, GitHub, receipt



Whiteboard + group check

Class → object → method → output. Predict the result in Console, WPF and the engine. Keep three concepts, one prediction, one question and lecture/whiteboard evidence. Groups name a lead and list tasks, owners and status. Each learner identifies their contribution.

Suggested class pacing

15 min lecture; 10 min board/team plan; 25 min Console; 35 min WPF; 50 min engine attempt/research/AI; 15 min Git/receipt/submission. Keep the scheduled break. Adjust support and approve documented access alternatives when needed.

Official references / checked October 28, 2026

- C# classes / Microsoft ↗
- C# Console App / Microsoft ↗
- WPF Hello World / Microsoft ↗
- Unity / scripts and attachment ↗
- Unity / Start callback ↗
- Godot / C# setup ↗
- Godot / nodes and scenes ↗
- Visual Studio / Git repository ↗
- Unity / version control ↗
- Godot / version control ↗
- Free flowchart tool / diagrams.net ↗